

## ASSIGNMENT

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

- **Native:** Built for one platform (Android or iOS). Offers the best performance and is used for high-performance apps like games or banking apps.
- **Hybrid:** Built with web technologies and runs inside a native app. Commonly used for simple apps with faster development.
- **Cross-platform:** Uses one codebase for Android and iOS (e.g., Flutter). Ideal for businesses that want to save time and cost while maintaining good performance.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

- The Flutter Ecosystem is a collection of tools, libraries, widgets, and packages that help developers build attractive, fast, and responsive mobile apps from a single codebase for Android and iOS.

3. What is the Software Development Life Cycle (SDLC), and why is it Important in building successful applications?

- The SDLC is a step-by-step process for developing software, including planning, designing, developing, testing, deploying, and maintaining an application. It helps ensure the app is high-quality, organized, and meets user needs.

4.How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

- Flutter is used during the development phase to build the app, while the SDLC provides the overall process from planning to maintenance. Together, they help teams create reliable, efficient, and user-friendly mobile applications.

5.Why should developers understand these concepts before starting an application project?

- Understanding these concepts helps developers choose the right development approach, build better applications efficiently, reduce errors, and deliver apps that meet user and business requirements.